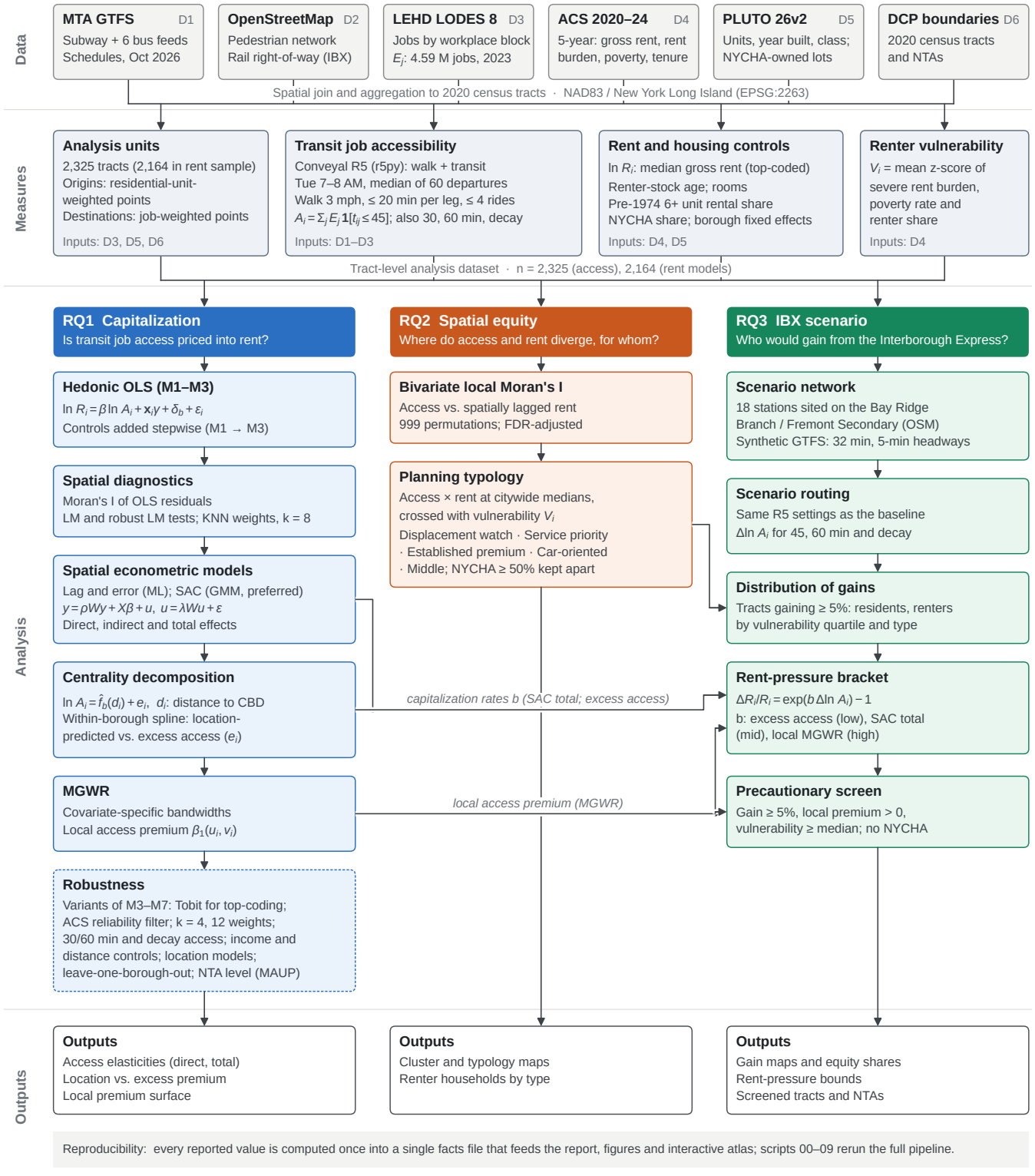
Data
MTA GTFS D1
Subway + 6 bus feeds
Schedules, Oct 2026
OpenStreetMap D2
Pedestrian network
Rail right-of-way (IBX)
LEHD LODES 8 D3
Jobs by workplace block
E_j: 4.59 M jobs, 2023
ACS 2020–24 D4
5-year: gross rent, rent burden, poverty, tenure
PLUTO 26v2 D5
Units, year built, class; NYCHA-owned lots
DCP boundaries D6
2020 census tracts and NTAs
Spatial join and aggregation to 2020 census tracts · NAD83 / New York Long Island (EPSG:2263)
Measures
Analysis units
2,325 tracts (2,164 in rent sample)
Origins: residential-unit-weighted points
Destinations: job-weighted points
Inputs: D3, D5, D6
Transit job accessibility
Conveyal R5 (r5py): walk + transit
Tue 7–8 AM, median of 60 departures
Walk 3 mph, ≤ 20 min per leg, ≤ 4 rides
$A_i = \Sigma_j E_j \mathbf{1}[t_{ij} \le 45]$; also 30, 60 min, decay
Inputs: D1–D3
Rent and housing controls
ln R_i: median gross rent (top-coded)
Renter-stock age; rooms
Pre-1974 6+ unit rental share
NYCHA share; borough fixed effects
Inputs: D4, D5
Renter vulnerability
V_i = mean z-score of severe rent burden, poverty rate and renter share
Inputs: D4
Tract-level analysis dataset · n = 2,325 (access), 2,164 (rent models)
Analysis
RQ1 Capitalization
Is transit job access priced into rent?
RQ2 Spatial equity
Where do access and rent diverge, for whom?
RQ3 IBX scenario
Who would gain from the Interborough Express?
Hedonic OLS (M1–M3)
$\ln R_i = \beta \ln A_i + \mathbf{x}_i\gamma + \delta_b + \varepsilon_i$
Controls added stepwise (M1 → M3)
Spatial diagnostics
Moran's I of OLS residuals
LM and robust LM tests; KNN weights, k = 8
Spatial econometric models
Lag and error (ML); SAC (GMM, preferred)
$y = \rho Wy + X\beta + u,\ u = \lambda Wu + \varepsilon$
Direct, indirect and total effects
Centrality decomposition
$\ln A_i = \hat{f}_b(d_i) + e_i,\ d_i$: distance to CBD
Within-borough spline: location-predicted vs. excess access (e_i)
MGWR
Covariate-specific bandwidths
Local access premium $\beta_1(u_i, v_i)$
Robustness
Variants of M3–M7: Tobit for top-coding; ACS reliability filter; k = 4, 12 weights; 30/60 min and decay access; income and distance controls; location models; leave-one-borough-out; NTA level (MAUP)
Bivariate local Moran's I
Access vs. spatially lagged rent
999 permutations; FDR-adjusted
Planning typology
Access × rent at citywide medians, crossed with vulnerability V_i
Displacement watch · Service priority · Established premium · Car-oriented · Middle; NYCHA ≥ 50% kept apart
Scenario network
18 stations sited on the Bay Ridge Branch / Fremont Secondary (OSM)
Synthetic GTFS: 32 min, 5-min headways
Scenario routing
Same R5 settings as the baseline
$\Delta \ln A_i$ for 45, 60 min and decay
Distribution of gains
Tracts gaining ≥ 5%: residents, renters by vulnerability quartile and type
capitalization rates b (SAC total; excess access)
Rent-pressure bracket
$\Delta R_i/R_i = \exp(b\,\Delta \ln A_i) - 1$
b: excess access (low), SAC total (mid), local MGWR (high)
local access premium (MGWR)
Precautionary screen
Gain ≥ 5%, local premium > 0, vulnerability ≥ median; no NYCHA
Outputs
Outputs
Access elasticities (direct, total)
Location vs. excess premium
Local premium surface
Outputs
Cluster and typology maps
Renter households by type
Outputs
Gain maps and equity shares
Rent-pressure bounds
Screened tracts and NTAs
Reproducibility: every reported value is computed once into a single facts file that feeds the report, figures and interactive atlas; scripts 00–09 rerun the full pipeline.